

LTI SYSTEM ANALYSIS

ENSC 429 – Digital Signal Processing

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LTI SYSTEM CHARACTERIZATION

- LTI systems are fully characterized by their impulse response $h[n]$ through the convolution equation

$$y[n] = h[n] * x[n]$$

- Taking DTFT of this equation

$$Y(e^{j\omega}) = H(e^{j\omega}) \cdot X(e^{j\omega})$$

- And taking z-transform

$$Y(z) = H(z) \cdot X(z)$$

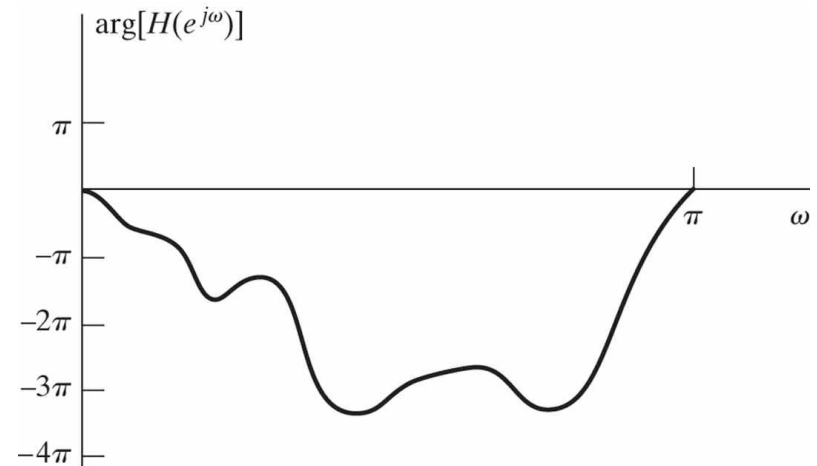
- Hence, an LTI system can also be fully characterized in the transform domain (DTFT or z-), via the transform of its impulse response
- $H(e^{j\omega})$ is called the **frequency response** of the system
- $H(z)$ is called the **system function**
- Both $H(e^{j\omega})$ and $H(z)$ are complex-valued, so we can represent them in rectangular form (real / imaginary parts), or polar form (magnitude / phase)

LTI SYSTEM CHARACTERIZATION

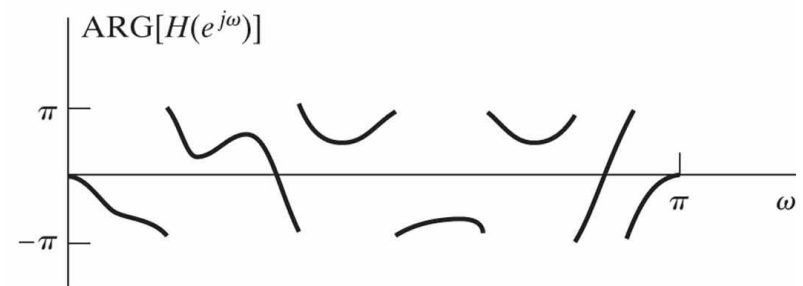
- In most cases, polar form is more useful
- For example, the frequency response in the polar form is

$$H(e^{j\omega}) = |H(e^{j\omega})| \angle H(e^{j\omega})$$

- $|H(e^{j\omega})|$ is called **magnitude response**
- $\angle H(e^{j\omega})$ is called **phase response**, sometimes also denoted $\arg[H(e^{j\omega})]$
- Phase of a complex number is not unique – adding an integer multiple of 2π to the phase will not change the complex number
- The principal value of the phase is denoted $\text{ARG}[H(e^{j\omega})]$ and is constrained to be in the interval $(-\pi, \pi]$



(a)



(b)

GROUP DELAY

- **Group delay** $\tau(\omega)$ is defined as the negative derivative of the phase

$$\tau(\omega) = \text{grd}[H(e^{j\omega})] = -\frac{d}{d\omega}\{\angle H(e^{j\omega})\}$$

- $\tau(\omega_0)$ is the delay experienced by the frequencies near ω_0
- Example: ideal delay system

$$h_{\text{id}}[n] = \delta[n - n_d]$$

- A LTI system with this impulse response would delay each input signal (all frequencies) by n_d samples; the frequency response is:

$$H(e^{j\omega}) = \text{DTFT}\{\delta[n - n_d]\} = e^{-j\omega n_d}$$

- So, the group delay is:

$$\tau(\omega) = -\frac{d}{d\omega}\{\angle H(e^{j\omega})\} = -\frac{d}{d\omega}\{-\omega n_d\} = n_d$$

- Hence, each frequency component is delayed by n_d

GROUP DELAY

- Another example: ideal lowpass filter with linear phase

$$H_{lp}(e^{j\omega}) = \begin{cases} e^{-j\omega n_d} & |\omega| < \omega_c \\ 0 & \omega_c < |\omega| \leq \pi \end{cases}$$

- The magnitude response is

$$|H_{lp}(e^{j\omega})| = \begin{cases} 1 & |\omega| < \omega_c \\ 0 & \omega_c < |\omega| \leq \pi \end{cases}$$

so the system acts as an ideal lowpass filter

- It can be shown ([see notes](#) for details) that the impulse response is:

$$h_{lp}[n] = \frac{\sin \omega_c (n - n_d)}{\pi (n - n_d)}$$

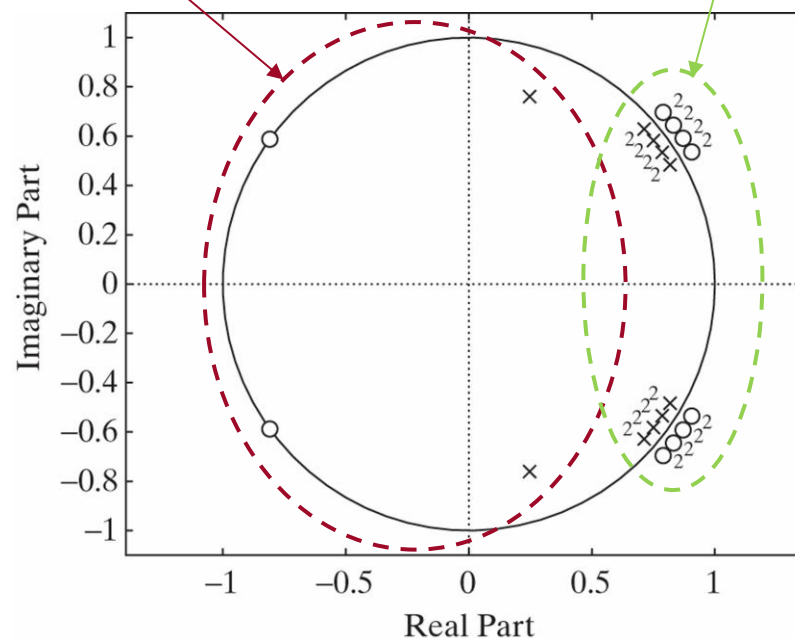
- Hence, compared to the ideal lowpass filter with zero phase, this impulse response is delayed by n_d samples

GROUP DELAY

- Another example: consider a LTI system with the following system function

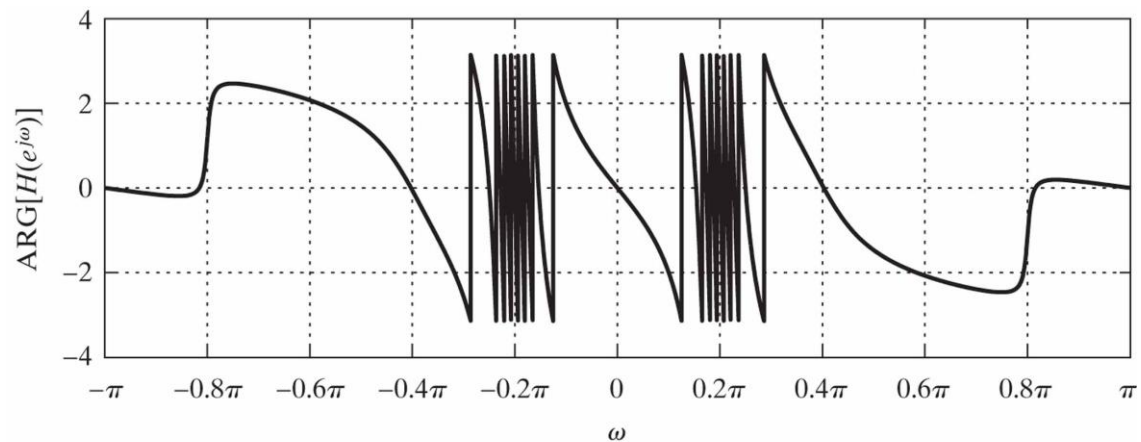
$$H(z) = \underbrace{\left(\frac{(1 - .98e^{j.8\pi} z^{-1})(1 - .98e^{-j.8\pi} z^{-1})}{(1 - .8e^{j.4\pi} z^{-1})(1 - .8e^{-j.4\pi} z^{-1})} \right)}_{H_1(z)} \underbrace{\prod_{k=1}^4 \left(\frac{(c_k^* - z^{-1})(c_k - z^{-1})}{(1 - c_k z^{-1})(1 - c_k^* z^{-1})} \right)^2}_{H_2(z)}$$

- Pole-zero diagram:

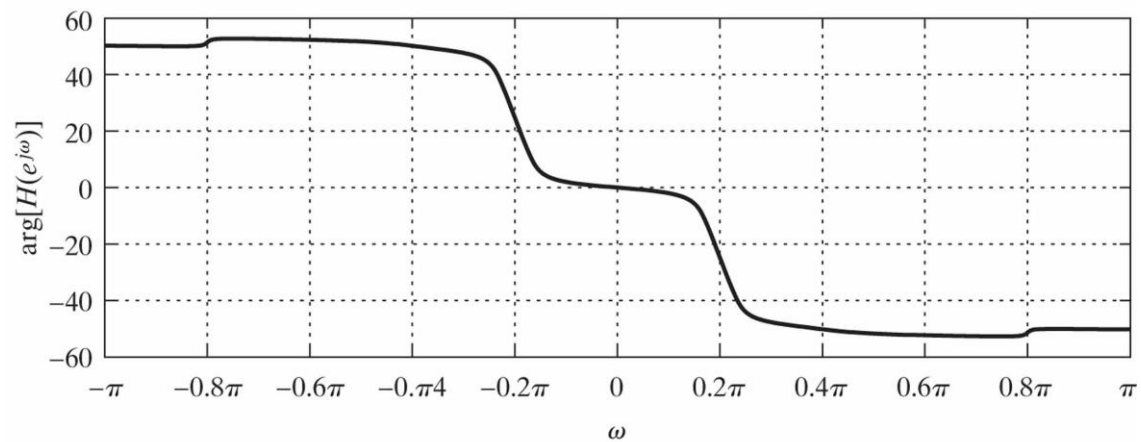


GROUP DELAY

Phase response



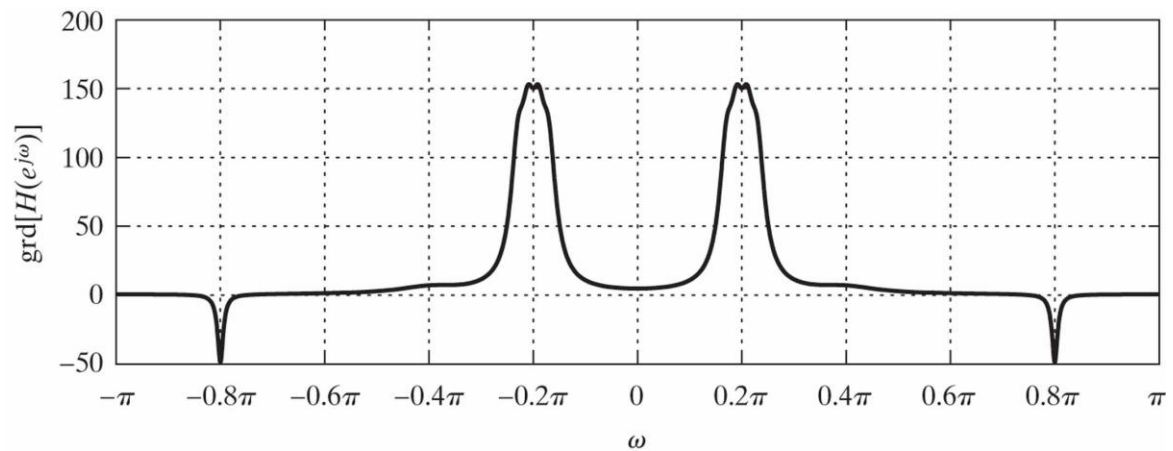
(a) Principle Value of Phase Response



(b) Unwrapped Phase Response

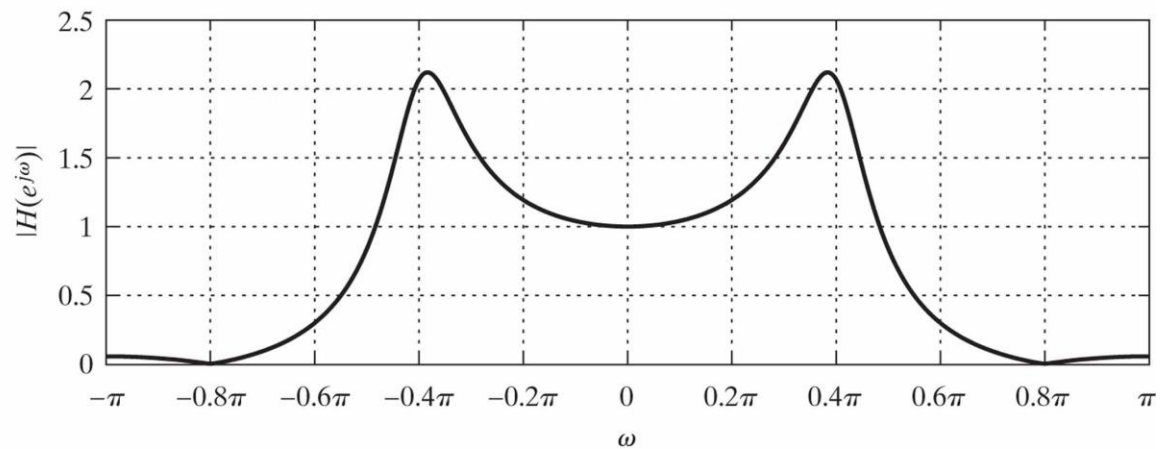
GROUP DELAY

Group delay



(a) Group delay of $H(z)$

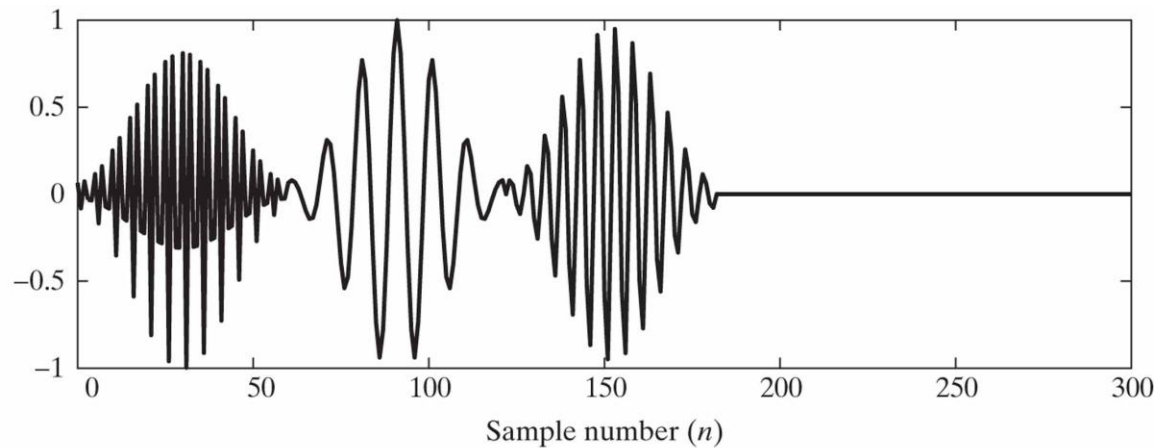
Magnitude response



(b) Magnitude of Frequency Response

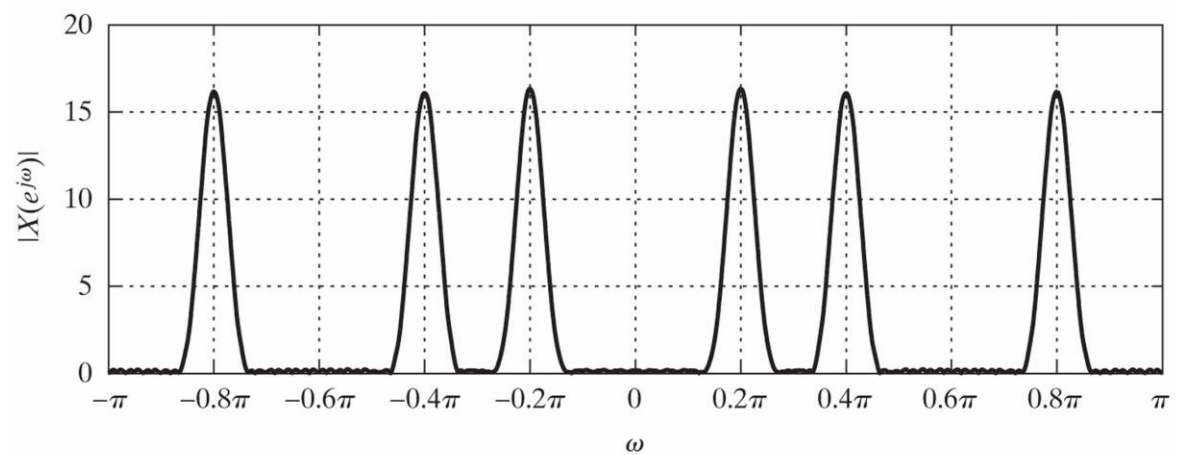
GROUP DELAY

Input signal $x[n]$



(a) Waveform of signal $x[n]$

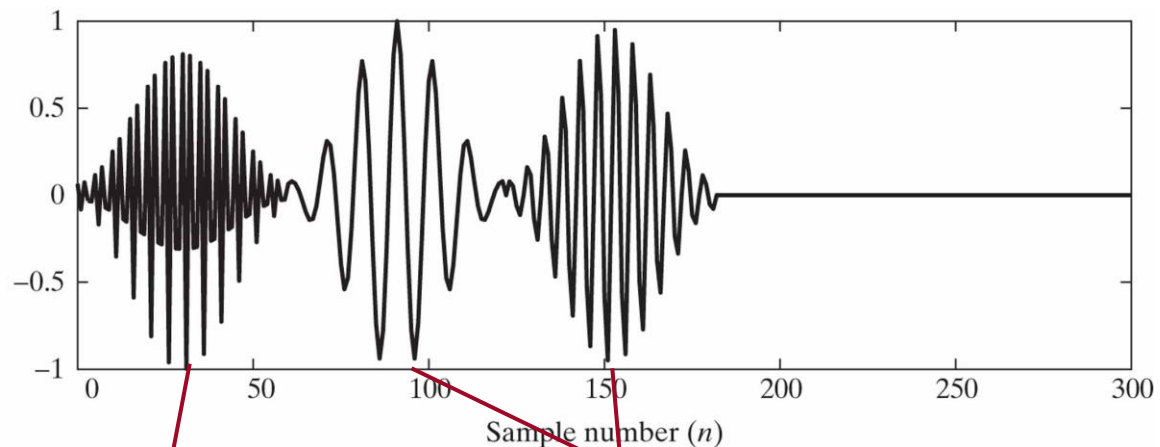
Its magnitude spectrum



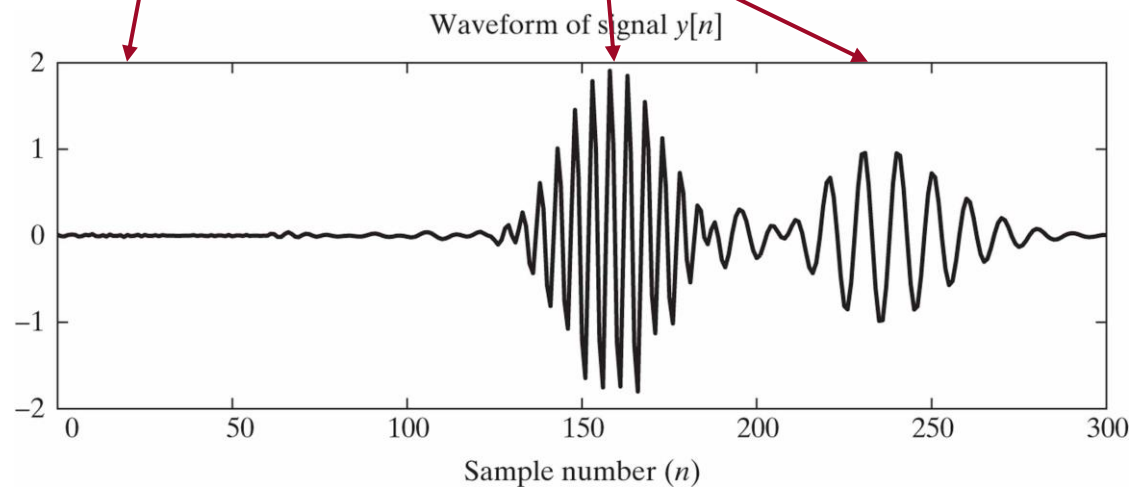
(b) Magnitude of DTFT of $x[n]$

GROUP DELAY

Input signal $x[n]$



Output signal $y[n]$



LTI SYSTEM AS A LCCDE

- LTI systems are often characterized (and implemented) as linear constant-coefficient difference equations (LCCDE)

$$a_0y[n] + a_1y[n-1] + \dots + a_Ny[n-N] = b_0x[n] + b_1x[n-1] + \dots + b_Mx[n-M]$$

- Applying the z-transform (remembering $\mathcal{Z}\{\delta[n-m]\} = z^{-m}$):

$$a_0Y(z) + a_1Y(z)z^{-1} + \dots + a_NY(z)z^{-N} = b_0X(z) + b_1X(z)z^{-1} + \dots + b_MX(z)z^{-M}$$

$$Y(z)[a_0 + a_1z^{-1} + \dots + a_Nz^{-N}] = X(z)[b_0 + b_1z^{-1} + \dots + b_Mz^{-M}]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1z^{-1} + \dots + b_Mz^{-M}}{a_0 + a_1z^{-1} + \dots + a_Nz^{-N}} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}}$$

- Numerator and denominator are polynomials in z^{-1} so, assuming no higher-order poles, they can be factored as

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 \prod_{k=1}^M (1 - c_k z^{-1})}{a_0 \prod_{k=1}^N (1 - d_k z^{-1})}$$

LTI SYSTEM AS A LCCDE

- Example: find the difference equation corresponding to the LTI system with the following system function:

$$H(z) = \frac{(1 + z^{-1})^2}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{3}{4}z^{-1}\right)}$$

- See notes for details

$$y[n] + \frac{1}{4}y[n-1] - \frac{3}{8}y[n-2] = x[n] + 2x[n-1] + x[n-2]$$

STABILITY AND CAUSALITY

- We saw that a given LCCDE can be mapped into a rational system function $H(z)$
- Depending on the poles and zeros, this system function can have different ROCs, and therefore correspond to different impulse responses $h[n]$
 - Remember: you need $H(z)$ and ROC to find $h[n]$
- Since a causal $h[n]$ is a right-sided sequence, from the properties of ROC we know that the **ROC of a causal system is outward from the outermost pole**
- Meanwhile, a stable system has an absolutely summable $h[n]$:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

- Note that this is the same condition as the existence of $H(z)$, for $|z| = 1$:

$$\sum_{n=-\infty}^{\infty} |h[n]z^{-n}| < \infty$$

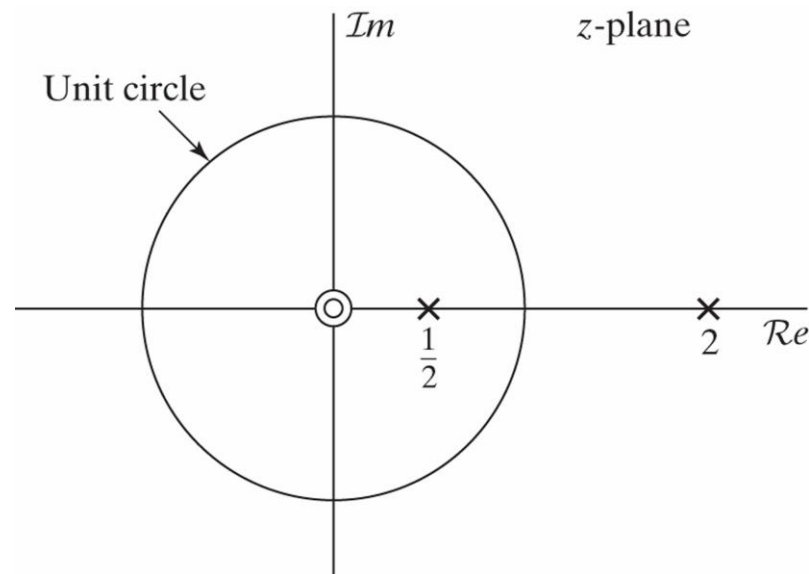
STABILITY AND CAUSALITY

- Therefore, **for the system to be stable, ROC of $H(z)$ must contain the unit circle**
- If the system is to be **both causal and stable**, the outermost pole must be within the unit circle, i.e., **all poles must be within unit circle**, and the ROC should be outside of the outermost pole
- Clearly, there could be systems that are causal but not stable, or stable but not causal, or neither, or both

- Example:

$$y[n] - \frac{5}{2}y[n-1] + y[n-2] = x[n]$$

- [See notes](#) for details



INVERSE SYSTEMS

- Sometimes we want to invert an action of a system
 - For example, a measurement process might distort the signal we want to measure, so we want to be able to “undistort” it
- If the distortion system is modeled by $H(z)$, the inverse system should have the system function $H_i(z)$ such that

$$H(z) \cdot H_i(z) = 1$$

- In other words, the inverse system should completely “undo” what $H(z)$ has done, so that when signal passes through both systems, it is as if nothing happened
- Therefore

$$H_i(z) = \frac{1}{H(z)}$$

- The above equations will be valid in the intersection of ROCs of $H(z)$ and $H_i(z)$
- For there to be a non-empty intersection, the two ROCs must have some overlap

INVERSE SYSTEMS

- If the distorter system function is

$$H(z) = \frac{b_0 \prod_{k=1}^M (1 - c_k z^{-1})}{a_0 \prod_{k=1}^N (1 - d_k z^{-1})}$$

then the inverse system function is

$$H_i(z) = \frac{1}{H(z)} = \frac{a_0 \prod_{k=1}^N (1 - d_k z^{-1})}{b_0 \prod_{k=1}^M (1 - c_k z^{-1})}$$

- Hence, the poles of $H(z)$ become the zeros of $H_i(z)$ and vice versa
- This exchange of poles and zeros can create interesting situations in terms of causality and stability of the inverse system
- We should always keep in mind that the ROCs of $H(z)$ and $H_i(z)$ must have some non-empty overlap

INVERSE SYSTEMS

- Example:

$$H(z) = \frac{1 - 0.5z^{-1}}{1 - 0.9z^{-1}}, \quad \text{ROC: } |z| > 0.9$$

- [See notes](#) for details
- The impulse response of the inverse system is:

$$h_i[n] = (0.5)^n u[n] - 0.9(0.5)^{n-1} u[n-1]$$

INVERSE SYSTEMS

- Example:

$$H(z) = \frac{z^{-1} - 0.5}{1 - 0.9z^{-1}}, \quad \text{ROC: } |z| > 0.9$$

- See notes for details
- Two possible impulse responses for the inverse system:

$$h_{i,1}[n] = 2(2)^n u[-n - 1] - 1.8(2)^{n-1} u[-n]$$

$$h_{i,2}[n] = -2(2)^n u[n] + 1.8(2)^{n-1} u[n - 1]$$

IMPULSE RESPONSE OF RATIONAL SYSTEM FUNCTIONS

- In the absence of higher-order poles, a rational system function can be expressed as

$$H(z) = \frac{b_0 \prod_{k=1}^M (1 - c_k z^{-1})}{a_0 \prod_{k=1}^N (1 - d_k z^{-1})} = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{\substack{k: \\ d_k \text{ first-order pole}}} \frac{A_k}{1 - d_k z^{-1}}$$

- If the system is causal, the ROC is outside the outermost pole: $|z| > \max |d_k|$
- With such ROC, the inverse z-transform gives the following impulse response:

$$h[n] = \sum_{r=0}^{M-N} B_r \delta[n - r] + \sum_{\substack{k: \\ d_k \text{ first-order pole}}} A_k (d_k)^n u[n]$$

- If the second sum exists (i.e., if there are some non-zero poles): **infinite (duration) impulse response (IIR)**
- If the second sum does not exist (i.e., if there are no non-zero poles): **finite (duration) impulse response (FIR)**

IMPULSE RESPONSE OF RATIONAL SYSTEM FUNCTIONS

- FIR system function is of the form

$$H(z) = \sum_{k=0}^M b_k z^{-k}$$

- Only has poles at $z = 0$, ROC is $|z| > 0$
- Taking the inverse z-transform:

$$h[n] = \sum_{k=0}^M b_k \delta[n - k]$$

- For the given input $x[n]$, the output is ([see notes](#) for details):

$$\begin{aligned} y[n] &= x[n] * h[n] = \sum_{m=-\infty}^{\infty} h[m] x[n - m] = \sum_{m=-\infty}^{\infty} \sum_{k=0}^M b_k \delta[m - k] x[n - m] \\ &= \sum_{k=0}^M b_k x[n - k] \end{aligned}$$

← Weighted linear combination of delayed inputs

IMPULSE RESPONSE OF RATIONAL SYSTEM FUNCTIONS

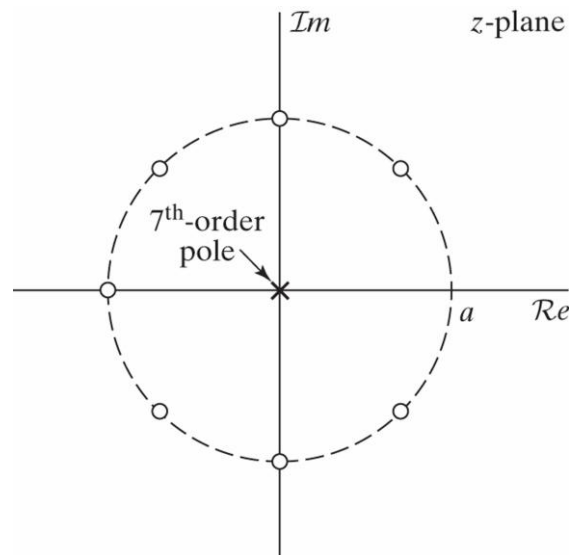
- Example: finite-duration exponential impulse response

$$h[n] = \begin{cases} a^n & 0 \leq n \leq M \\ 0 & \text{else} \end{cases}$$

- There are two difference equations that could implement this system:

$$y[n] = \sum_{k=0}^M a^k x[n-k], \quad \text{or} \quad y[n] - ay[n-1] = x[n] - a^{M+1}x[n-M-1]$$

- [See notes](#) for details



FREQUENCY RESPONSE OF RATIONAL SYSTEM FUNCTIONS

- It is useful to be able to sketch the frequency response, especially magnitude response, for a given system function $H(z)$
- Consider a simple system function with $a = re^{j\theta}$:

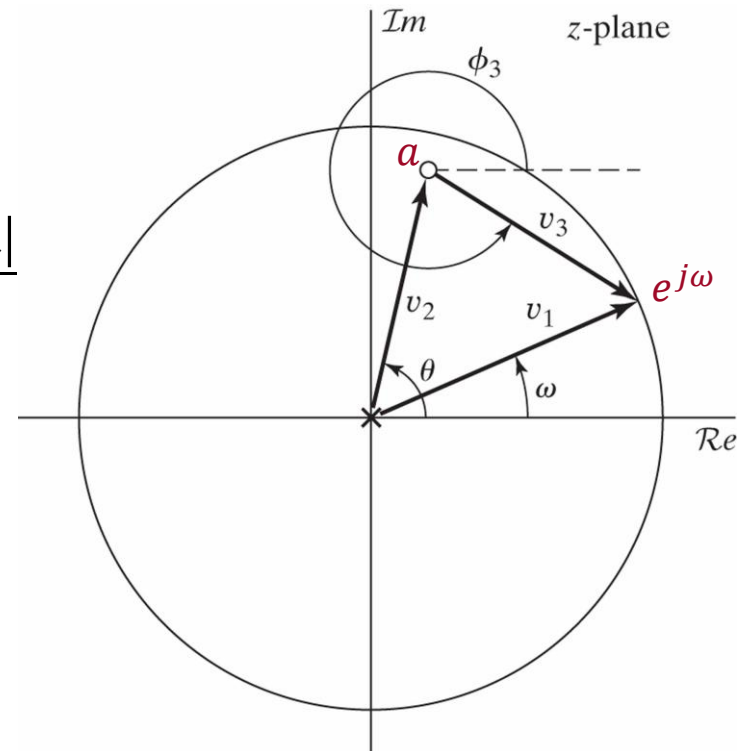
$$H(z) = 1 - az^{-1} = \frac{z - a}{z}$$

- Magnitude response:

$$\begin{aligned} |H(e^{j\omega})| &= |H(z)|_{z=e^{j\omega}} = \left| \frac{e^{j\omega} - a}{e^{j\omega}} \right| = \frac{|e^{j\omega} - a|}{|e^{j\omega}|} \\ &= \frac{|v_3|}{|v_1|} = \frac{|v_3|}{1} = |v_3| \end{aligned}$$

- Phase response:

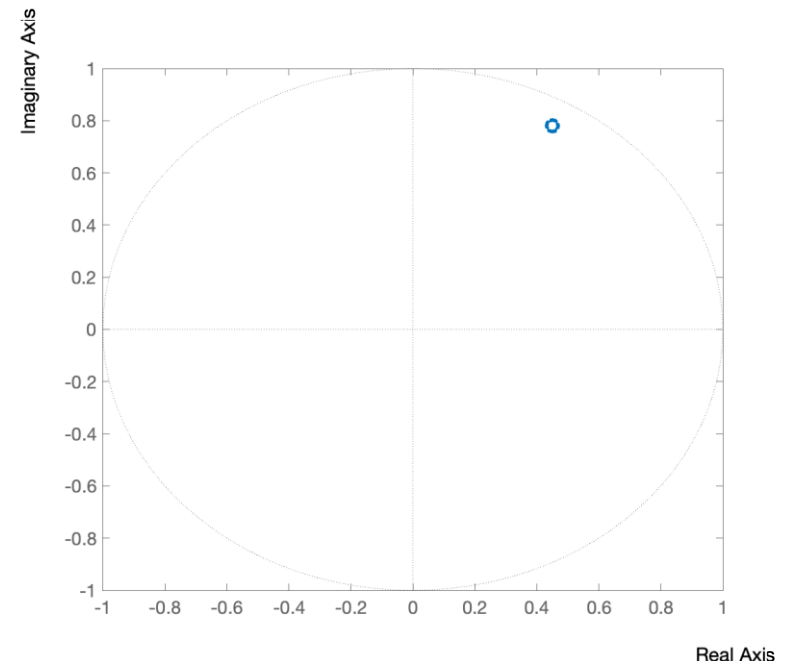
$$\begin{aligned} \angle H(e^{j\omega}) &= \angle H(z) \Big|_{z=e^{j\omega}} = \angle \left(\frac{e^{j\omega} - a}{e^{j\omega}} \right) \\ &= \angle(e^{j\omega} - a) - \angle(e^{j\omega}) = \phi_3 - \omega \end{aligned}$$



FREQUENCY RESPONSE OF RATIONAL SYSTEM FUNCTIONS

MATLAB function `pzmap` plots the pole-zero map of the system function

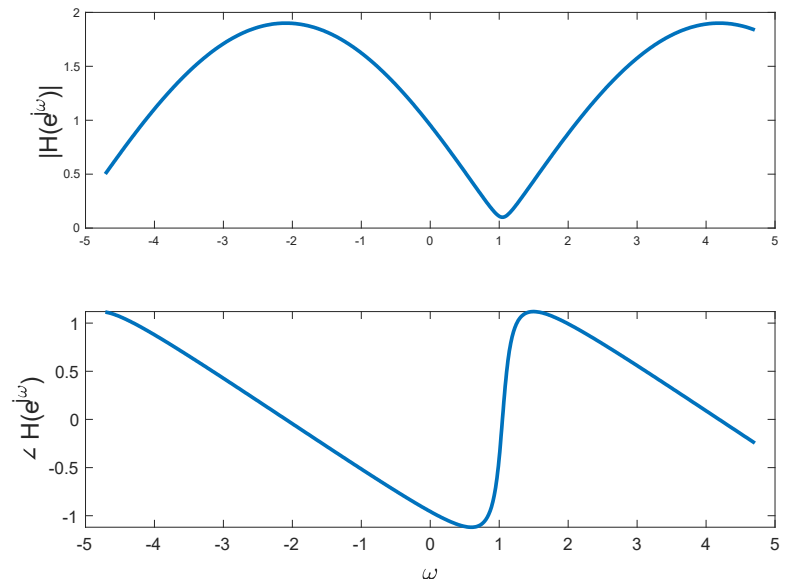
```
r = 0.9; theta = pi/3;  
a = r*exp(j*theta);  
num = [1 -a]; % numerator coefficients: 1 - az^(-1)  
den = [1]; % denominator coefficients: 1  
sys = tf(num,den,0.1);  
pzmap(sys);
```



FREQUENCY RESPONSE OF RATIONAL SYSTEM FUNCTIONS

MATLAB function `freqz` computes the DT frequency response

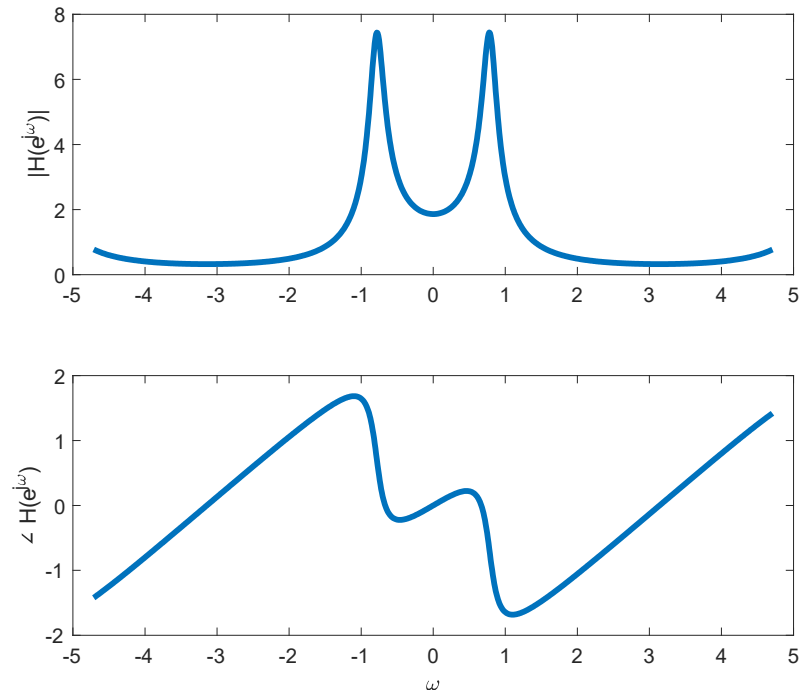
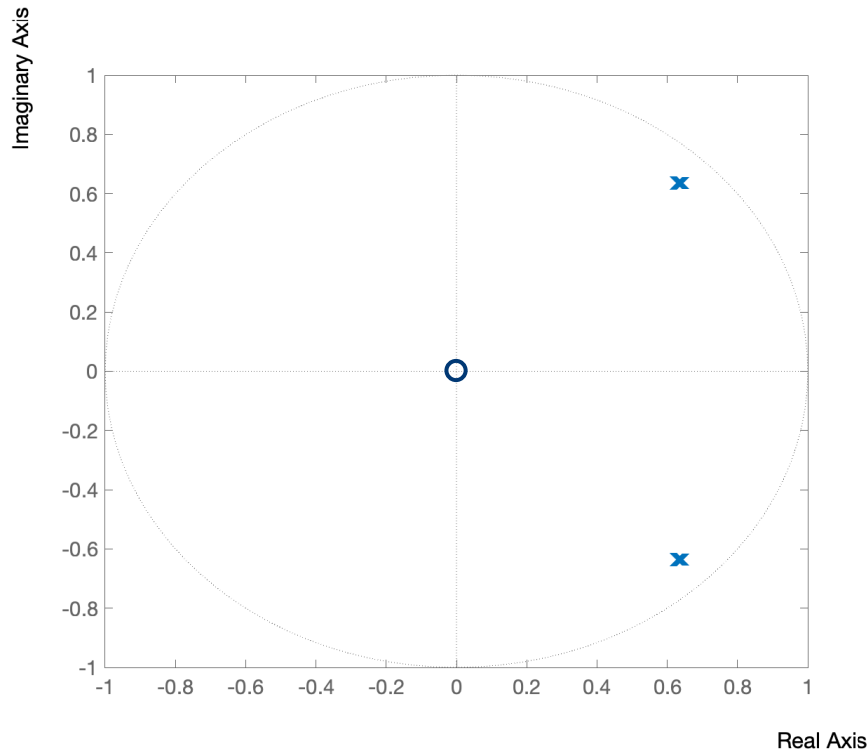
```
r = 0.9; theta = pi/3;  
a = r*exp(j*theta);  
num = [1 -a]; % numerator coefficients: 1 - az^(-1)  
den = [1]; % denominator coefficients: 1  
omega = -1.5*pi:0.01:1.5*pi; % frequencies  
H = freqz(num,den,omega);  
subplot(2,1,1);  
plot(omega,abs(H)); grid  
ylabel('|H(e^{j\omega})|');  
subplot(2,1,2);  
plot(omega,angle(H)); grid  
xlabel('\omega');  
ylabel('\angle H(e^{j\omega})');
```



FREQUENCY RESPONSE OF RATIONAL SYSTEM FUNCTIONS

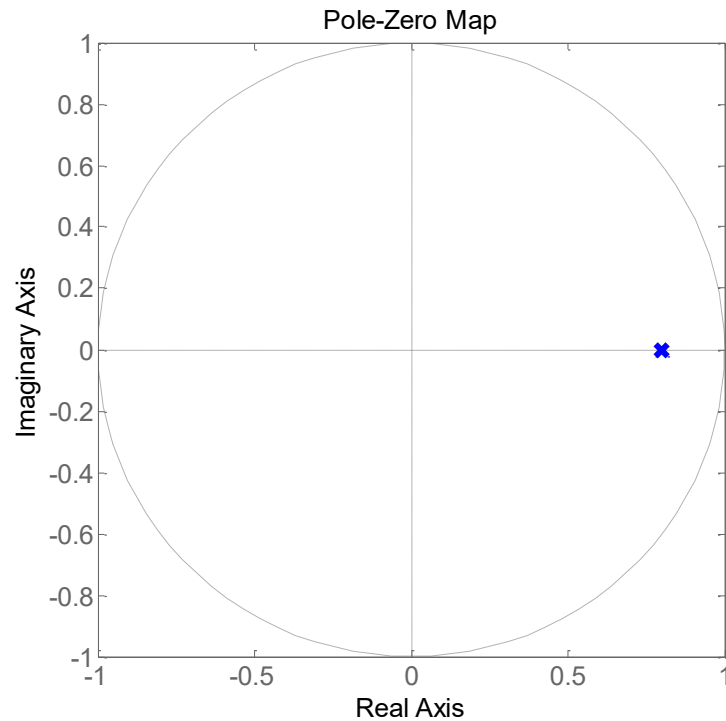
- Example: let $a = 0.9e^{j\pi/4}$ and consider this system function ([see notes](#))

$$H(z) = \frac{1}{(1 - az^{-1})(1 - a^*z^{-1})} = \frac{1}{1 - 1.2728z^{-1} + 0.81z^{-2}}$$

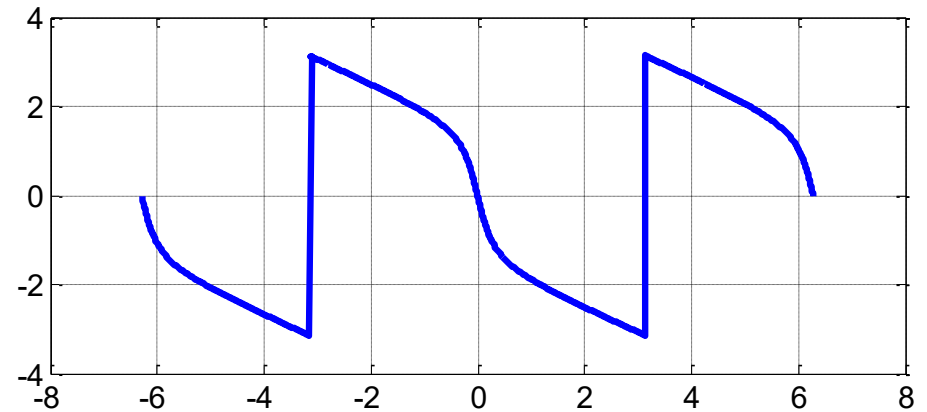
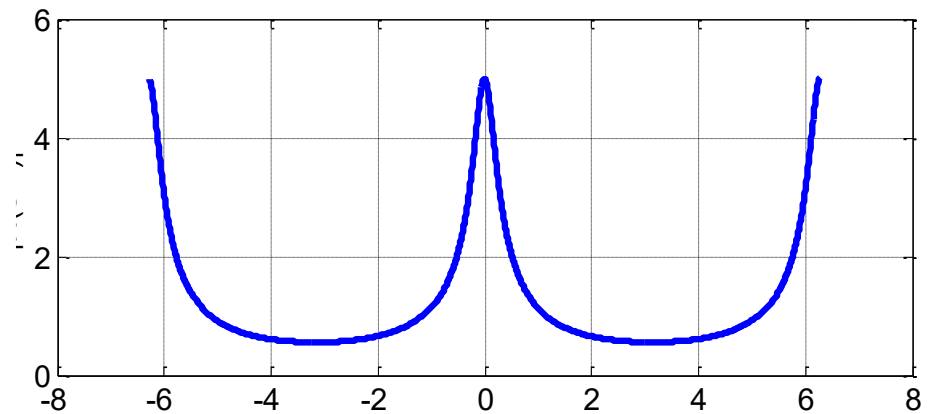


EXAMPLE 1

$$H(z) = \frac{1}{z - 0.8}$$

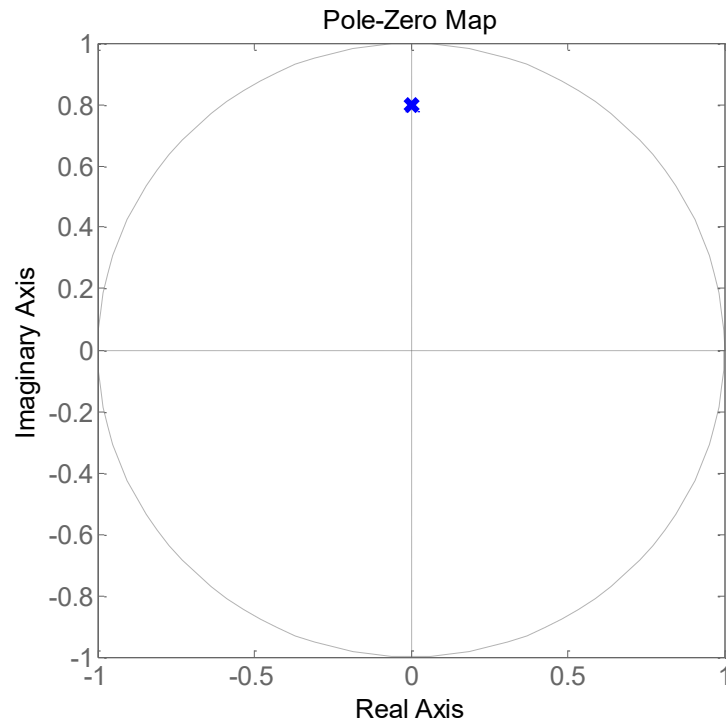


Low-pass filter

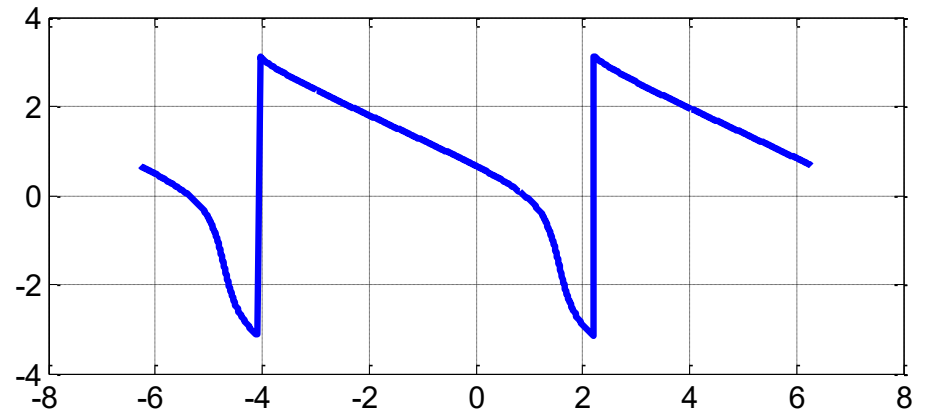
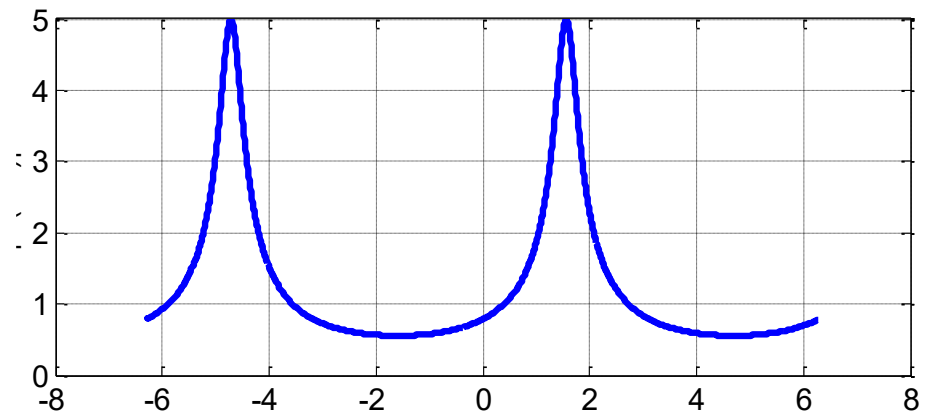


EXAMPLE 2

$$H(z) = \frac{1}{z - j0.8}$$

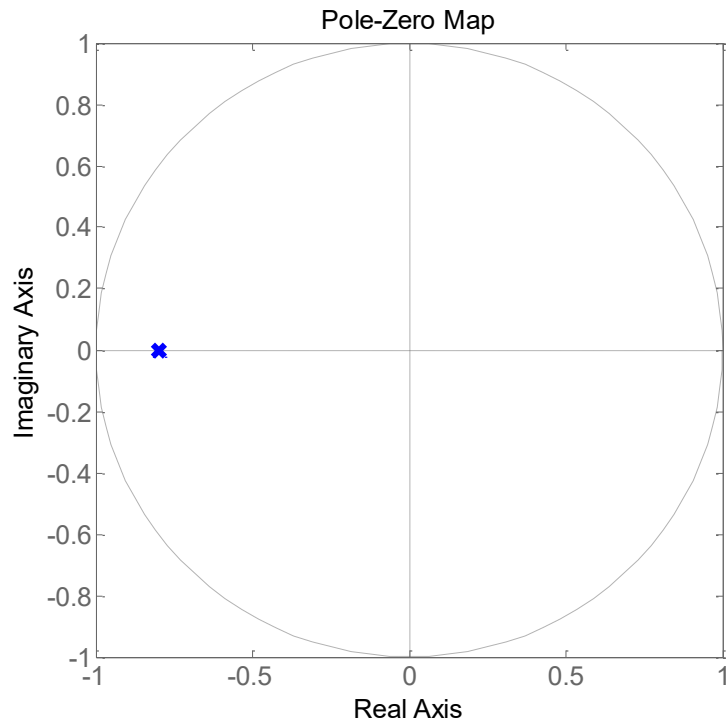


Band-pass filter

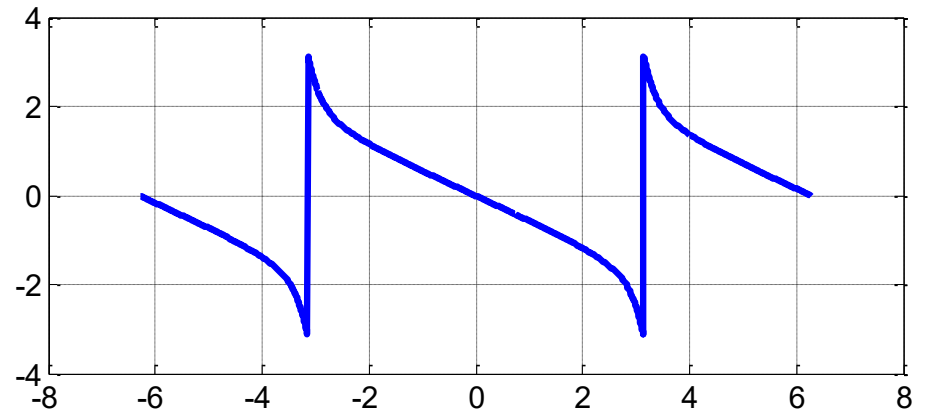
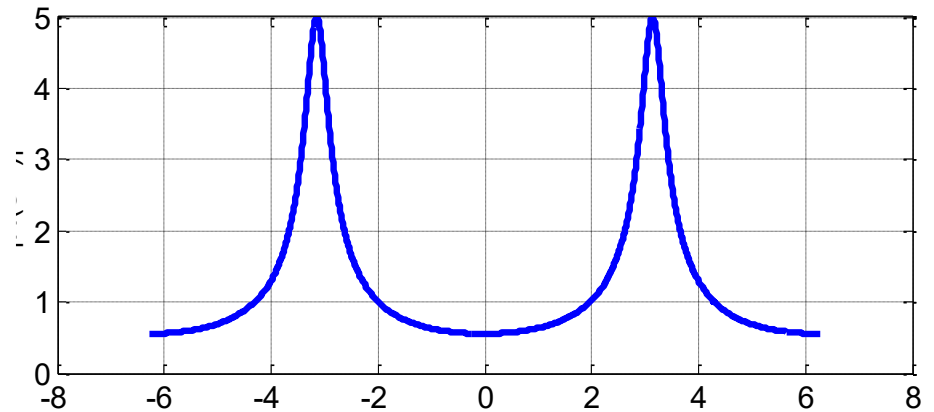


EXAMPLE 3

$$H(z) = \frac{1}{z + 0.8}$$

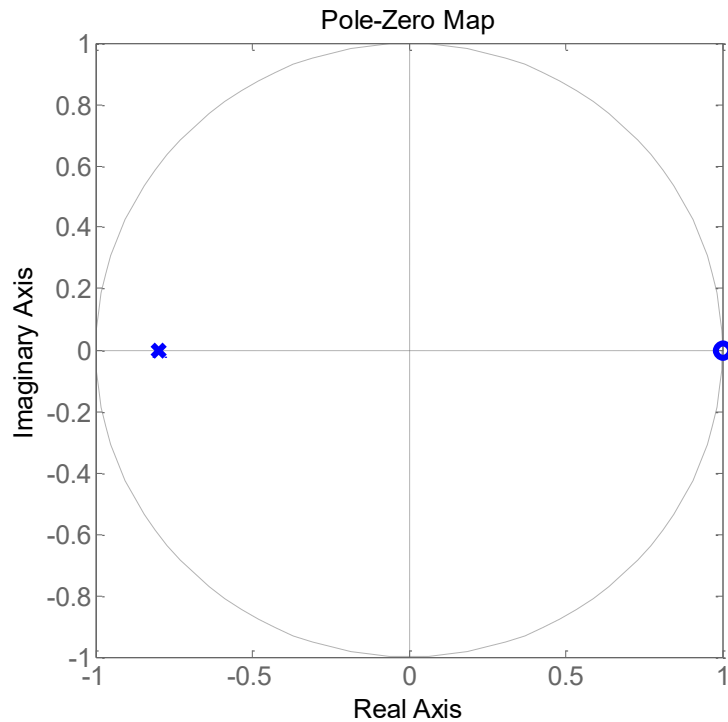


High-pass filter

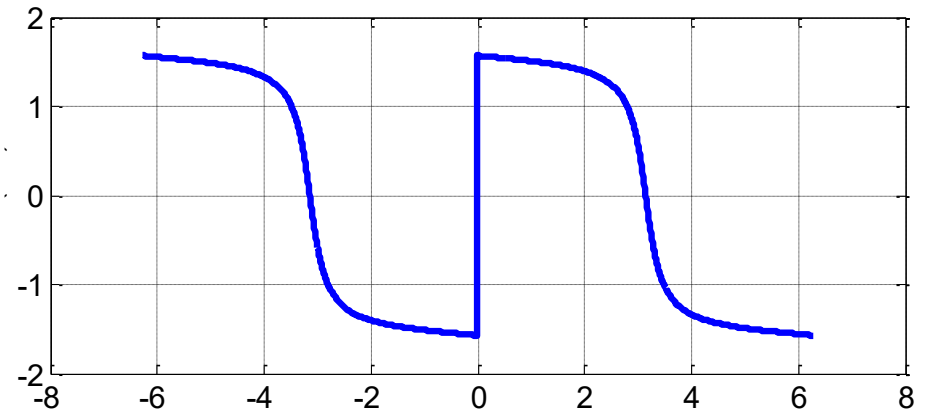
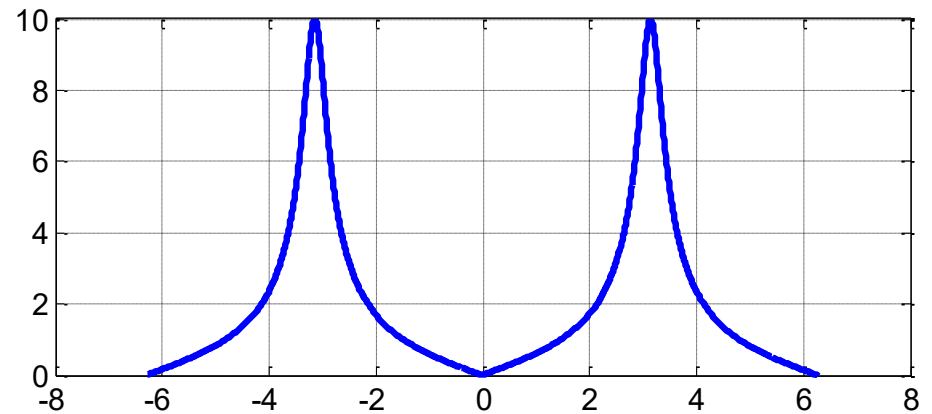


EXAMPLE 4

$$H(z) = \frac{z - 1}{z + 0.8}$$



High-pass filter



READING ASSIGNMENT

- Read Sections 5.1 – 5.3 in the textbook
- Start reading Chapter 6